Physics 336
HW9

Solve the following problems. Do at least one in Mathematica, and at least one in Python.

1. Recall that in a series RLC circuit, the impedance is expected to follow

$$Z = \sqrt{R^2 + \left(2\pi f L - \frac{1}{2\pi f C}\right)^2}$$

The resistor voltage, $V_R = IR = (V_0/Z)R$ can be measured over a range of frequencies to determine the values of L, and C. Suppose the frequency is adjusted in 50-Hz increments, and V_R is measured:

$$f(\text{Hz}) = \{50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750\}$$
$$V_R(\text{V}) = \{0.40, 0.83, 1.20, 1.49, 1.67, 1.74, 1.74, 1.68, 1.60, 1.51, 1.42, 1.34, 1.26, 1.18, 1.11\}$$

Suppose we measure $V_0 = 1.80$ V and $R = 17.0\ \Omega$ using a multimeter. Analyze these data to determine values for L and C.

2. The activity of a mixture of *two* radioisotopes is measured using a Geiger counter, with counts measured in forty 10-second intervals, centered one-minute apart (beginning at 1.00 min)

$$R(\text{counts}) = \{2343, 2066, 1839, 1690, 1490, 1397, 1232, 1171, 1017, 1000, 956, 864, 795, 793, 748, 736, 703, 662, 636, 622, 564, 567, 541, 512, 534, 497, 504, 456, 466, 469, 434, 410, 421, 400, 413, 380, 348, 376, 341, 344\}$$

Fit this data to an appropriate function, and interpret the results.

3. A portion of the solar spectrum near an absorption line has been measured.

$$\lambda(\text{nm}) = \{740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780\} \pm 1$$
$$I(\text{arbitrary units}) = \{0.277, 0.275, 0.273, 0.271, 0.269, 0.266, 0.263, 0.259, 0.256, 0.252, 0.248, 0.244, 0.240, 0.237, 0.232, 0.222, 0.202, 0.179, 0.159, 0.138, 0.120, 0.105, 0.099, 0.107, 0.119, 0.131, 0.143, 0.154, 0.161, 0.166, 0.166, 0.165, 0.163, 0.160, 0.158, 0.155, 0.152, 0.149, 0.146, 0.144, 0.141\} \pm 0.003$$

Treat this as a Gaussian *absorption* with a background, and analyze this data.

4. A distant star is a binary system. It is imaged with a telescope, and the counts per pixel are measured along a line through the center of the image.

   pixel, $p = \{22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43\}$
   counts, $c = \{1, 3, 7, 14, 26, 43, 64, 84, 99, 106, 102, 92, 82, 77, 75, 70, 56, 37, 19, 8, 3, 1\}$

   Assuming the light from each star creates a different Gaussian distribution (with unknown amplitude A, mean μ, and standard deviation σ) across pixels, fit this data to find the relative brightness of these two stars.

5. A converging lens is used to cast real images on a screen. The object and image distances are recorded.

   $$\begin{aligned} d_o(\mathrm{cm}) =& \{18.0, 21.0, 24.0, 27.0, 30.0, 40.0, 46.0, 50.0, 53.0\} \ \pm 0.1 \,\text{each} \\ d_i(\mathrm{cm}) =& \{101 \pm 3, 56 \pm 2, 43.0 \pm 1.5, 35.5 \pm 0.7, 31.1 \pm 0.6, 25.2 \pm 0.4, \\ & 23.1 \pm 0.3, 22.2 \pm 0.3, 21.9 \pm 0.3\} \end{aligned}$$

   Plot d_i *vs.* d_o, and fit it it to an appropriate function. Use this to determine the power (in diopters) of the lens.